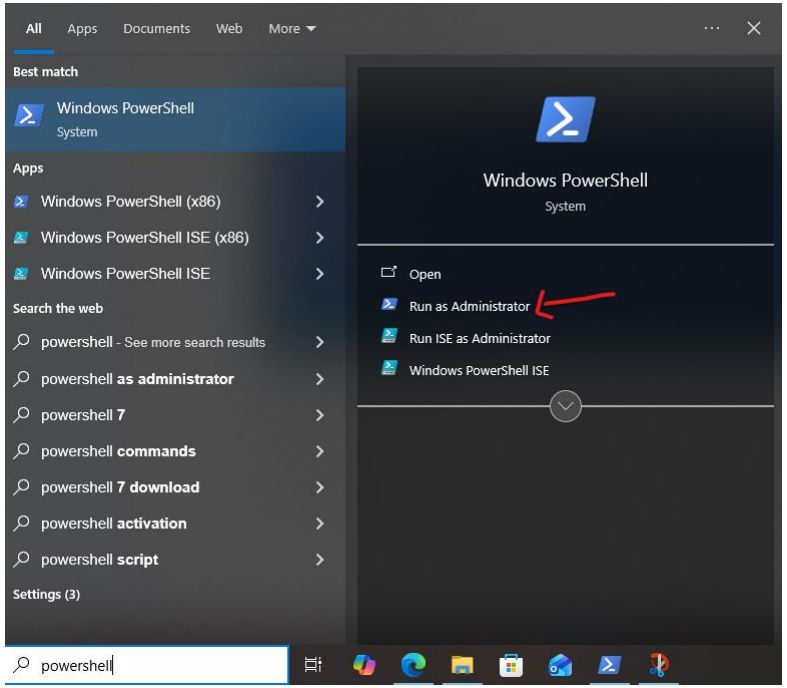
Linux/WSL

WSL Installation

Step 1: Open Windows PowerShell as an administrator, and approve the subsequent User Account Control prompt.



Step 2: Execute *wsl --install -d Ubuntu-24.04*. By default, the distribution will be installed on the C: drive. The download requires approximately 1.5 GB of space and will take an estimated 2 minutes to complete.

Step 2 : wsl --install -d Ubuntu-24.04 # By default it will download in C drive , download will take around 2 min. (1.5 Gb of space)

Step 3 : create username and password

```
durg@DESKTOP-2AA7GHT: /mnt/c/WINDOWS/system32
PS C:\WINDOWS\system32>  wsl --install -d Ubuntu-24.04
Downloading: Ubuntu 24.04 LTS
Installing: Ubuntu 24.04 LTS
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu-24.04'
Launching Ubuntu-24.04...
Provisioning the new WSL instance Ubuntu-24.04
This might take a while...
Create a default Unix user account: durg
New password:
Retype new password:
passwd: password updated successfully
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$
```

If you need to move to different drive - let's say moving to E drive

Step 4 : exit # come out of linux

Step 5 : mkdir E:\WSL

Step 6 : wsl --shutdown

Step 7 : wsl --export Ubuntu-24.04 E:\WSL\ubuntu-backup.tar

Step 8 : wsl --unregister Ubuntu-24.04 #to unregister the c drive version

Step 9 : wsl --import Ubuntu-24.04 E:\WSL\Ubuntu_Distro E:\WSL\ubuntu-backup.tar #Importing to E drive

Step 10 : optional : rm E:\WSL\ubuntu-backup.tar

```
durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ exit
exit
PS C:\WINDOWS\system32>  mkdir E:\WSL


    Directory: E:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        26-03-2026     15:15                WSL


PS C:\WINDOWS\system32>  wsl --shutdown
PS C:\WINDOWS\system32> wsl --export Ubuntu-24.04 E:\WSL\ubuntu-backup.tar
Export in progress, this may take a few minutes. (1314 MB)

The operation completed successfully.
PS C:\WINDOWS\system32> wsl --unregister Ubuntu-24.04
Unregistering.
The operation completed successfully.

PS C:\WINDOWS\system32>  wsl --import Ubuntu-24.04 E:\WSL\Ubuntu_Distro E:\WSL\ubuntu-backup.tar
The operation completed successfully.
PS C:\WINDOWS\system32>  rm E:\WSL\ubuntu-backup.tar
PS C:\WINDOWS\system32>
```

Step 11 : wsl # To enter the linux system

```
durg@DESKTOP-2AA7GHT: /mnt/c/WINDOWS/system32
logout
PS C:\WINDOWS\system32> wsl
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$
```

Conda Installation

#Whenever prompted Proceed ([y]/n)? Always enter y.

mkdir -p ~/miniconda3
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -O ~/miniconda3/miniconda.sh
bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3
rm -rf ~/miniconda3/miniconda.sh
echo 'export PATH=~/miniconda3/bin:$PATH' >> ~/.bashrc && source ~/.bashrc

```
durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ mkdir -p ~/miniconda3
durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ rm -r ~/miniconda3
durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ exit
logout
PS C:\WINDOWS\system32> wsl
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ mkdir -p ~/miniconda3
ps://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -O ~/miniconda3/miniconda.sh

bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3

rm -rf ~/miniconda3/miniconda.sh

echo 'export PATH=~/miniconda3/bin:$PATH' >> ~/.bashrc && source ~/.bashrc

durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$
durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -O ~/miniconda3/miniconda.sh
--2026-03-26 09:58:07--  https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
Resolving repo.anaconda.com (repo.anaconda.com)... 104.16.191.158, 104.16.32.241, 2606:4700::6810:20f1, ...
Connecting to repo.anaconda.com (repo.anaconda.com)|104.16.191.158|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 162067098 (155M) [application/octet-stream]
Saving to: ‘/home/durg/miniconda3/miniconda.sh’

/home/durg/miniconda3/minicon 100%[===================================================>] 154.56M  28.8MB/s    in 5.9s

2026-03-26 09:58:13 (26.2 MB/s) - ‘/home/durg/miniconda3/miniconda.sh’ saved [162067098/162067098]

durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$
durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3
PREFIX=/home/durg/miniconda3
Unpacking bootstrapper...
Unpacking payload...

Installing base environment...

Preparing transaction: ...working... done
Executing transaction: ...working... done
installation finished.
durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$
durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ rm -rf ~/miniconda3/miniconda.sh
durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$
durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ echo 'export PATH=~/miniconda3/bin:$PATH' >> ~/.bashrc && source ~/.bashrcdurg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$
```

We will downgrade condas for stable version

conda install -c conda-forge conda=25.7.0 #accept the Terms of Conditions whenever needed , and give permission for downloading whenever asked

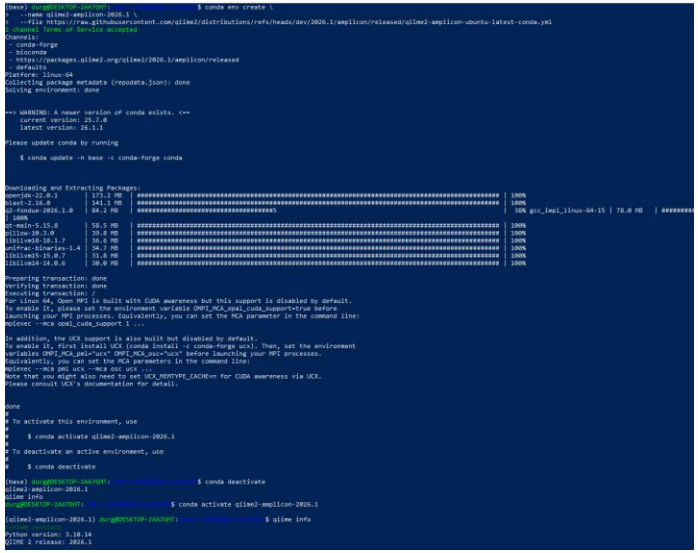
conda init
exit
wsl

```
durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ conda install -c conda-forge conda=25.7.0
2 channel Terms of Service accepted
Retrieving notices: done
Channels:
 - conda-forge
 - defaults
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: /home/durg/miniconda3

  added / updated specs:
    - conda=25.7.0


The following packages will be downloaded:

    package                    |            build
    ---------------------------|-----------------
    ca-certificates-2026.2.25  |       hbd8a1cb_0         144 KB  conda-forge
    certifi-2026.2.25          |     pyhd8ed1ab_0         148 KB  conda-forge
    conda-25.7.0               |  py313h78bf25f_0         1.2 MB  conda-forge
    conda-libmamba-solver-25.4.0|     pyhd8ed1ab_0          41 KB  conda-forge
    openssl-3.6.1              |       h35e630c_1         3.0 MB  conda-forge
    ------------------------------------------------------------
                                           Total:         4.5 MB

The following packages will be UPDATED:

  ca-certificates    pkgs/main/linux-64::ca-certificates-2~ --> conda-forge/noarch::ca-certificates-2026.2.25-hbd8a1cb_0
  certifi            pkgs/main/linux-64::certifi-2026.01.0~ --> conda-forge/noarch::certifi-2026.2.25-pyhd8ed1ab_0
  openssl               pkgs/main::openssl-3.5.5-h1b28b03_0 --> conda-forge::openssl-3.6.1-h35e630c_1

The following packages will be SUPERSEDED by a higher-priority channel:

  conda              pkgs/main::conda-26.1.1-py313h06a4308~ --> conda-forge::conda-25.7.0-py313h78bf25f_0
  conda-libmamba-so~ pkgs/main::conda-libmamba-solver-25.1~ --> conda-forge::conda-libmamba-solver-25.4.0-pyhd8ed1ab_0


Proceed ([y]/n)? y


Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$
```

QIIME 2 Installation (Estimated time: 15 minutes)

```
conda env create \
  --name qiime2-amplicon-2026.1 \
  --file https://raw.githubusercontent.com/qiime2/distributions/refs/heads/dev/2026.1/amplicon/released/qiime2-amplicon-ubuntu-latest-conda.yml
```

```
#To test the installation
conda deactivate
conda activate qiime2-amplicon-2026.1
qiime info
```



QC Installation (multiqc)

```
# Create a new environment named 'qc' with Python
```

conda create -n qc python=3.12

```
(qiime2-amplicon-2026.1) durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ conda create -n qc python=3.12
2 channel Terms of Service accepted
Channels:
 - defaults
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: /home/durg/miniconda3/envs/qc

  added / updated specs:
    - python=3.12
```

Activate the environment
conda activate qc

Install MultiQC from the Bioconda channel
conda install -c conda-forge -c bioconda multiqc

```
(qiime2-amplicon-2026.1) durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ conda activate qc
(qc) durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ conda install -c conda-forge -c bioconda multiqc
2 channel Terms of Service accepted
Channels:
 - conda-forge
 - bioconda
 - defaults
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done
```

Trimmomatic Installation

conda create -n trimmo

conda activate trimmo

conda install -c bioconda trimmomatic

```
durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ conda create -n trimmo
2 channel Terms of Service accepted
Channels:
 - defaults
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: /home/durg/miniconda3/envs/trimmo


Proceed ([y]/n)? y


Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate trimmo
#
# To deactivate an active environment, use
#
#     $ conda deactivate

durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ conda activate trimmo
(trimmo) durg@DESKTOP-2AA7GHT:/mnt/c/WINDOWS/system32$ conda install -c bioconda trimmomatic
2 channel Terms of Service accepted
Channels:
 - bioconda
 - defaults
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done
```

Mac